Predict Students' Dropout and Academic Success

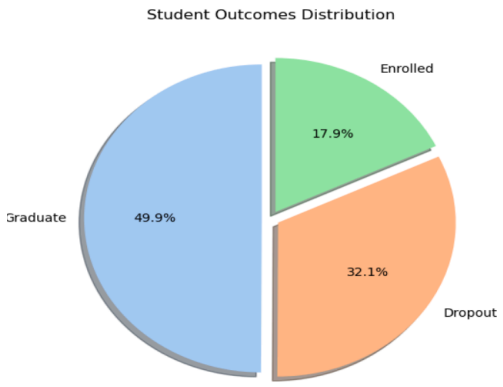
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Abstract

Student dropout has been a big challenge for educational institutions. It affects academic performance and long-term outcomes. By predicting at-risk students early, schools can help them stay on track as soon as they need. This research is conducted by using machine learning techniques, demographic, socioeconomic, and academic performance data. Three main models, which are Support Vector Machine (SVM), Multi-layer Perceptron classifier, and Neural Network, are used to train and evaluate the data. The accuracy and F1-score of each model are used to compare model performance. Top 15 important features affecting academic performance will be found to identify the strongest predictors of dropout risk.

1. Introduction

College success is viewed by many as absolutely crucial, and in some occupations, it is vital for working in one's field of study. It can be devastating to fail for anybody, and sometimes it can be hard to pinpoint precisely what caused a student to drop out. This study provides a way to identify the variables that are common with those likely to drop out. For this study, we utilize the "Predict Students' Dropout and Academic Success" dataset created by M.V. Martins, D. Tolledo, J. Machado, L.M.T. Baptista, and V. Realinho. The dataset contains information regarding students enrollment in different undergraduate degrees, such as design, education, nursing, journalism, management, social service, and technology. This dataset contains 4000 instances, where each instance is a student. It also contains 36 features such as tuition, parental education, marital status, age at enrollment, courses, and nationality.



The target variable represents a student's final academic status at the end of their degree program:

- Dropout: Student left the program before completing it.
- Enrolled: Student is still studying but has not graduated.
- Graduate: Student completed all requirements and finished the program.

Since we mainly focus on the dropout and non-dropout rates, graduate and enrolled rates are combined into the non-dropout rate.

We chose this topic because we wanted to help reduce academic dropout and failure in higher education. To do that, we need to analyze this dataset, build predictive models, and find the top factors affecting academic performance.

2. Related work

Given the increasing use of machine learning in educational data mining, there is plenty of helpful related work from which we were able to refer for this project. For example, in Predicting school failure using data mining, [1], the researchers used traditional machine learning algorithms and tree-based ensemble classifiers to predict dropout among university students. After comparing the two techniques, they found that tree-based ensemble models performed better with margins of 8%–12% points in accuracy.

Another relevant reference is Predictors of higher education dropout intention in the post-pandemic era: The mediating role of academic exhaustion. It states that after the predictive model was implemented, "academic exhaustion was the stronger predictor, followed by social connection to the campus, vocational difficulties, and course value." [2]

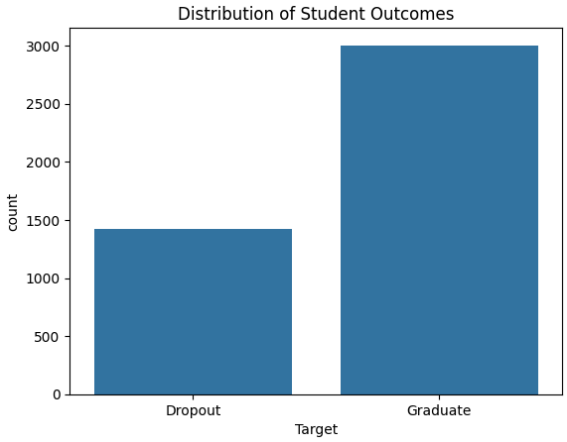
3. Proposed Method

In this project, we implemented 3 models, which are Support Vector Machine (SVM), Multi-layer Perceptron classifier (MLP), and Neural Network (NN), to evaluate the academic performance of students. The chosen models must be widely used in educational data and be able to handle nonlinearity.

3.1. Data Preprocessing

Data preprocessing should include the handling of missing values, feature scaling, making the target variables numerical, and addressing class imbalance. As we mentioned previously, graduate and

enrollment rates are combined into the non-dropout rate.



The dataset is split into training and testing sets with a ratio of 80-20. Model performance is evaluated using accuracy and F1 Score.

3.2. Support Vector Machine (SVM)

Support Vector Machine (SVM) is widely used for classification and regression. It uses a hyperplane to separate different classes in the data. SVMs can perform multi-class or binary classification, which is useful for separating nondropout and dropout data into two classes.

3.3. Multi-layer Perceptron Classifier (MLP)

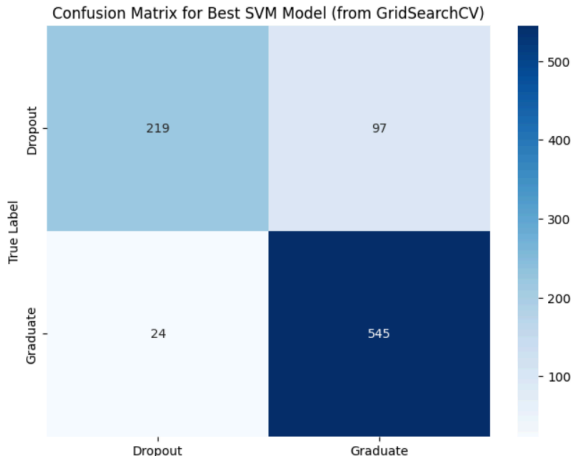
A Multi-layer Perceptron (MLP) classifier is used to classify an input into the category to which it belongs. There is an input layer, one or more hidden layers, and an output layer. Noticeably, all neurons in one layer are connected to all neurons in the next one. The purpose of an MLP is to model complex relationships between inputs and outputs, which matches the model requirements of this project.

3.4. Artificial Neural Network (ANN)

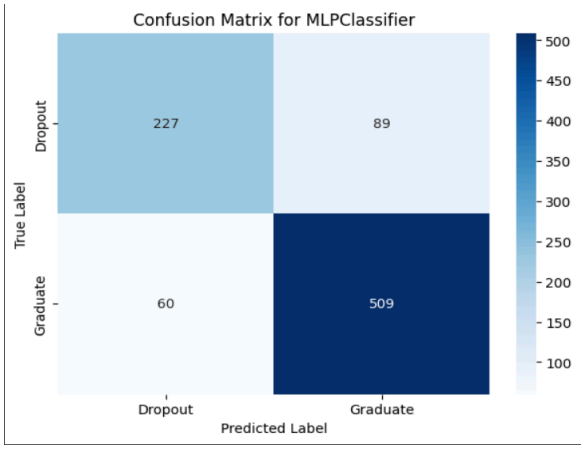
Besides the MLP classifier, we also implemented an Artificial Neural Network (ANN) model to compare performance with different neural network architectures. One of the benefits of ANN is the flexibility in the number of hidden layers. It allows experimentation with wider and deeper structures to observe changes in accuracy. The ANN architecture begins with an input layer, followed by hidden layers with ReLU activation. Dropout and L2 weight penalties are used for the output layer to reduce overfitting. During ANN training, we used the cross-entropy loss function and Adam optimizer, which allows the model to capture high-level nonlinear relationships.

4. Experiments

The first method used was an SVM using best-fit parameters by utilizing SVC (support vector classification). It was found that SVM would give an accuracy of 86% with an F-1 score of 85%.

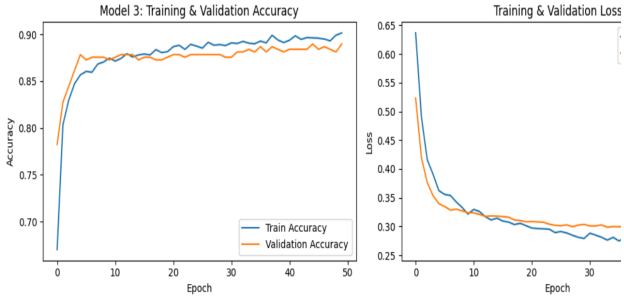


Overall decent, but they're were still other models to test. Moving on to MLP (multi-layer perceptron), we built an MLP model using the x_train_scaled data with the MLP classifier in the sklearn neural network library. Notable details include that it was built with three hidden layers (100, 50, 25) with a maximum of 500 iterations, an Adam optimizer, and its activation function is ReLU. After fitting the MLP classifier, it achieved 83% accuracy and an F1 score of 81%.

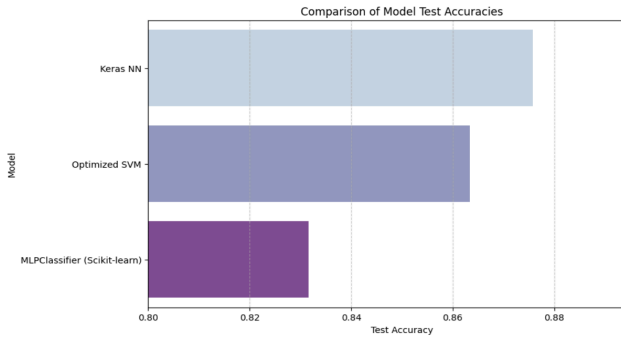


This model performed worse in terms of accuracy, making it unlikely to be chosen as the final model for this project. Finally, the custom multi-layer neural network model was tested. As it is a custom-made neural network, it has many unique things that were added throughout its making; however, starting off, there were three dense layers with the parameters of 64 to 32 to 16, all with ReLU activation functions, and then one output layer with it's loss being binary cross entropy, optimizer being adam, with it's metric only including accuracy. When running this model with 50 epochs, the accuracy came back as a high 90%, but there was a slight issue found. When using a validation split of 10% it was found that the validation accuracy would reach extreme lows (10%) while the validation loss would reach huge heights (89%). Thus, it indicated the model was overfitting or relying too much on the original trained data given that it made extremely off predictions when it came to the validation data. To solve this, dropout layers were added after the first two hidden layers, with a dropout rate of 30%. This would solve it by occasionally throwing out data and training without it, ensuring it wouldn't be too reliant on the training data. With this change, the custom multi-layer neural network model's overall

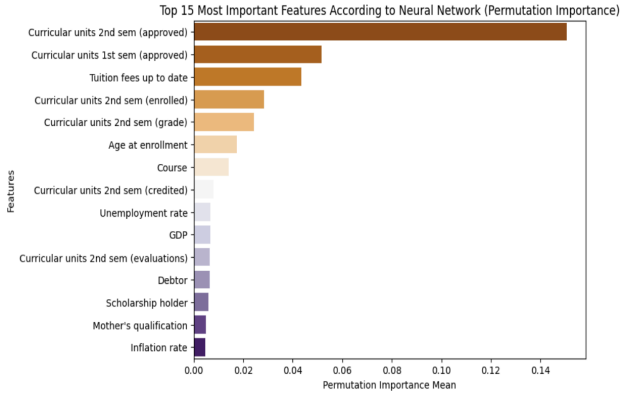
accuracy dropped to 88%, but it aligned the validation data with the training data, so it was considered an improvement overall.



5. Results



As seen in the picture above, the test accuracy shows that the best model to use would be the custom-built multi-layer neural network. Now that that model had been chosen, it was essential to determine which features of the dataset were most impactful on the model after training. In this pursuit, permutation importance was chosen as the method in which to determine feature importance. Permutation importance evaluates the importance of features by randomly shuffling the values or weights in a model to see how big of a drop in accuracy occurs for those particular features. After running a permutation importance model based on the custom multi-layer neural network and a general scoring method (if above 0.5, graduate; otherwise, dropout), it identified the top 15 most important features according to permutation importance.



It is very important to note that even if 0.14 away from zero seems like a small number, it is huge when talking about permutation importance. The most notable features are the top five, as all the other features generally hover around a 0.005 mean; therefore, they are equally as relevant/important to the target as each other. Firstly, the most crucial feature, with a permutation mean of 0.15, dwarfing many of the other features, is the number of units or courses approved for the second semester. Speculating on why this particular feature could be so crucial for academic success could have many reasons, such as students who have no courses approved for the next

semester, it would be generally assumed they had dropped out, as it differs from the typical path in college, basically going to college means astudent is trying to get a degree as fast as possible so he or she may use said degree earlier to get a career. Now, students' circumstances are all unique; some may need to take a semester off due to a personal issue, but, as this permutation importance shows, if a student is not devoting enough time to college, it hurts their chances of graduating. This nicely leads into the second most important feature (permutation importance mean of 0.05): the number of courses approved in the first semester, which further shows that if a student isn't dedicated time-wise, it hurts their chances of earning their degree. Thirdly comes the amount of tuition fees up to date (end of 2nd semester), with a mean of 0.04. It's easy to speculate on why it's so important on the model, as having too high of fees that you cannot handle will, in one way or another, force you to become a dropout, especially since the dataset is students at the end of their first year. At the end of the top five most important features lie the 2nd semester's amount of units enrolled (mean: 0.03) and grade (mean: 0.02). Enrolled kinda leads back to the most important feature of approved units, however, in this case, it is probably lower in importance, speculatively because the number of courses approved naturally lines up with the amount enrolled, therefore, by itself, won't cause too much change in accuracy, but still accounts for situations where a student approves multiple courses then doesn't follow through with enrolling them. Grades being an important feature is likely due to the fact that even if a student takes as many courses as they can without the necessary effort to succeed in those units/classes, their path for academic success can only lead to failure, and naturally, having low grades will only hurt their chances at graduation, regardless of any other features.

6. Conclusion

The goal of this project is not to say that if a student is taking college part-time, has any tuition fees, or has a rough first year in terms of grades, that they are destined to become a dropout, and that nothing they do matters. Vice versa, don't go overboard as it's just as likely to backfire, like taking too many courses that you can't handle, or reducing your tuition fees to zero by taking loans from loan sharks. Its true purpose is to serve as a warning to those concerned that their time in college is being wasted by failure,

the hope being that a student who identifies with this model sees that they are on a path to academic dropout and tries to change other aspects of their academic environment. That rough first year of college cannot be changed; it's in the past, and it's entirely possible a student will need to drop out because of how deep a hole they've dug, sometimes their fault, sometimes unavoidable circumstances. But whether one needs to start again or can rebound from that year, it is the hope that this early warning can allow a student to claw their way back to academic success. In terms of future direction for this project, adding more instances or students to the dataset could help flesh out the model and make it much better for the general student. Also, allowing for more personalized feature importance, such as removing all degrees other than a bachelor's degree in the features category, would anything change? Another thing that could be added in a future rendition of this project is some of today's more topical features. An especially topical feature of today is AI usage; it could be extremely interesting to see whether it will overall hurt chances of academic success. A different direction would be to either improve the custom model itself or switch to a superior model, as while many models and parameters, such as epochs, were tested, it's entirely possible that a better model exists. Overall, there are multiple directions in which this project can be improved and made more efficient, which can hopefully be made in the future.

7. References

[1] Márquez, Carlos & Romero, Cristóbal & Ventura, Sebastian. (2011). Predicting School Failure Using Data Mining.. EDM. 271-276.

[2] Gonzalez, B., Mendes, T. P., Pinto, R., Correia, S. V., Albuquerque, S., & Paulino, P. (2025). Predictors of higher education dropout intention in the post-pandemic era: The mediating role of academic exhaustion. *PLOS ONE, 20*(7), e0327643. https://doi.org/10.1371/journal.pone.0327643

[3] Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). *Predict Students' Dropout and Academic Success* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5MC89